Lecture 7

COMP 3760

Data Structures and Graphs

Text chapter 1.4, 3.5, 5.3

Fundamental Data Structures

(Chapter 1.4)

Data Structures

- A *data structure* is a particular way of storing and organizing data
- Data structures and algorithms are often deeply interconnected
  - The way you organize data affects the performance of your algorithm
- We've *mostly* been using arrays … so far

Fundamental Data Structures

- Linear Data Structures
  - Array
  - Linked list
  - Stack
  - Queue
- Set
- Dictionary (Map)
- Tree
- Graph

Arrays

- A sequence of *n* items of the same type, accessed by an index

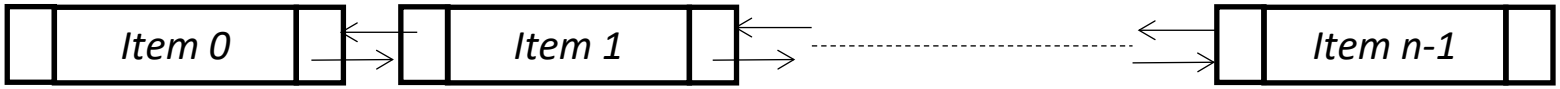
Item[0]	*Item[1]*	…	*Item[n-1]*

- The good:
  - Each item accessed in same constant time
- The bad:
  - Size is fixed
  - Insertion / deletion in an array is time consuming – all the elements following the inserted element must be shifted appropriately

Linked Lists

- (singly) A sequence of zero or more elements called *nodes,* consisting of data and a pointer

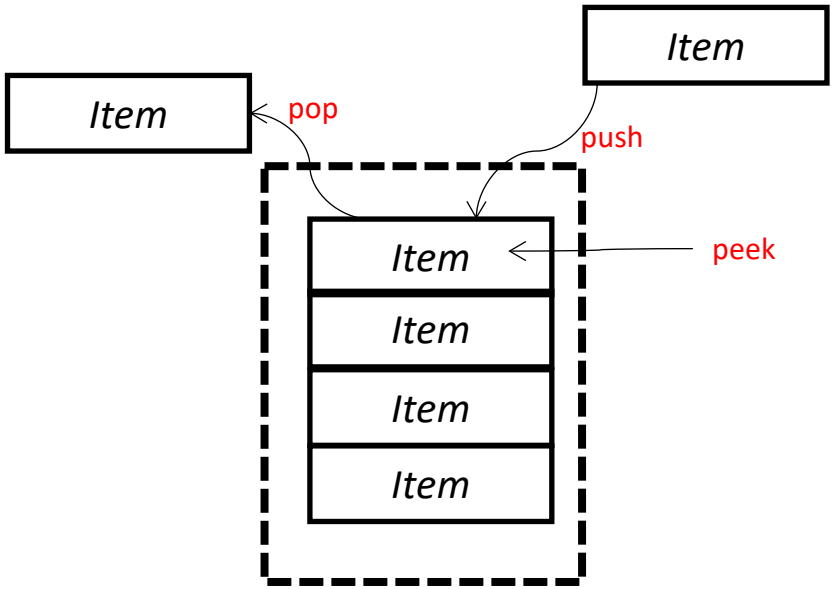
- (doubly) Pointers in each direction



Linked Lists

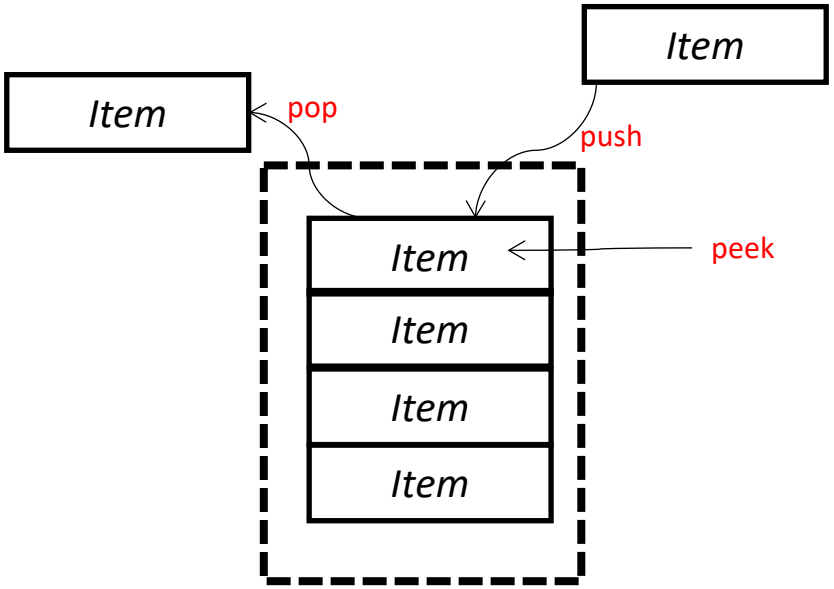
- Linked lists provide two key advantages over arrays
  - Dynamic size
  - Ease of insertion/deletion
- Linked lists have some drawbacks:
  - Random access is not allowed

Do you know what this is?



Stack

- Like a stack of plates
- Last-in-first-out (LIFO)



Operations on a stack

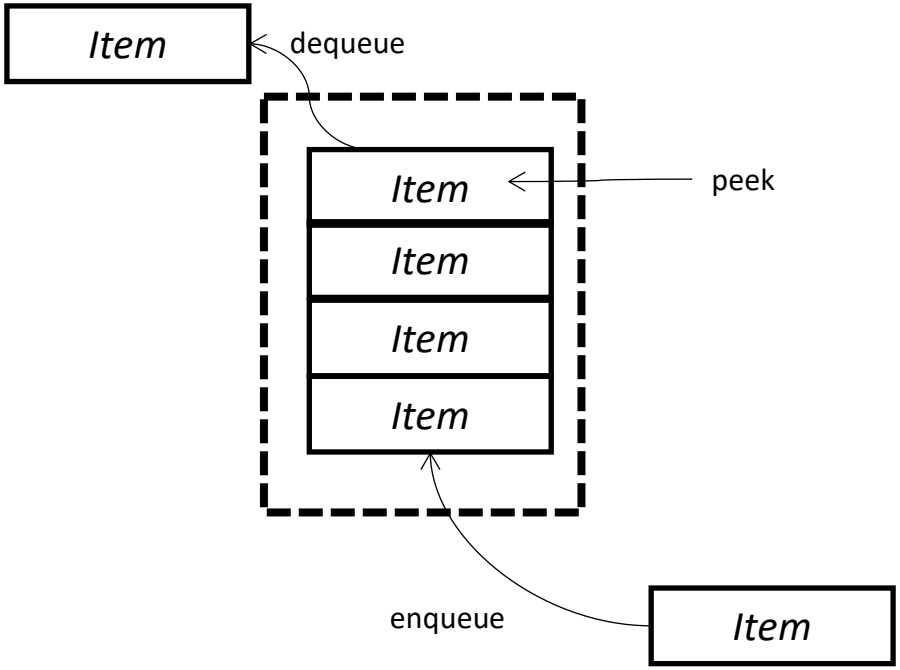
- Insert operation is called <u>Push</u>
- Delete operation is called <u>Pop</u>
- Examining the top item is <u>Peek</u>

- Example application:
  - Analysis of languages (e.g. properly nested brackets)
  - Properly nested: (())
  - Wrongly nested: ())(()

Abstract Data Type

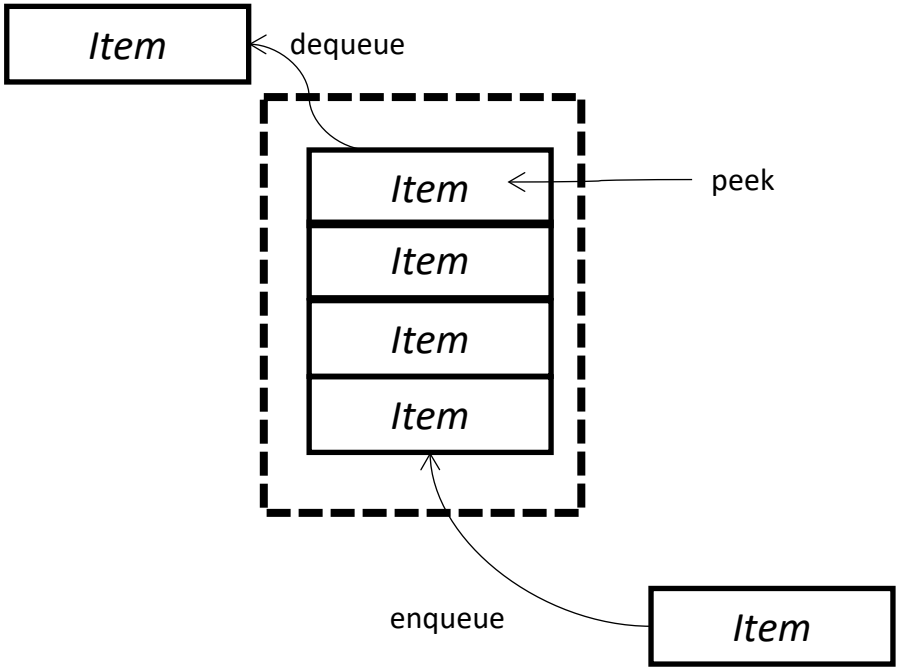
- Often a data structure is closely associated with a set of available operations
- **Data structure + operations = *abstract data type***
  - From an OOP perspective, think about members (methods) of a class
- Example 1: priority queue
  - Underlying implementation was a max-heap
  - Operations were Insert and deleteMax
- Example 2: stack
  - Operations are push, pop, peek

How about this one?



Queues

- Like a line-up
- First-in-first-out (FIFO)



Operations on a queue

- Adding to the queue is <u>Enqueue</u>
- Removing from the queue is <u>Dequeue</u>
- The top/front element is the <u>Head</u> (sometimes there is a “Peek” method)

Set

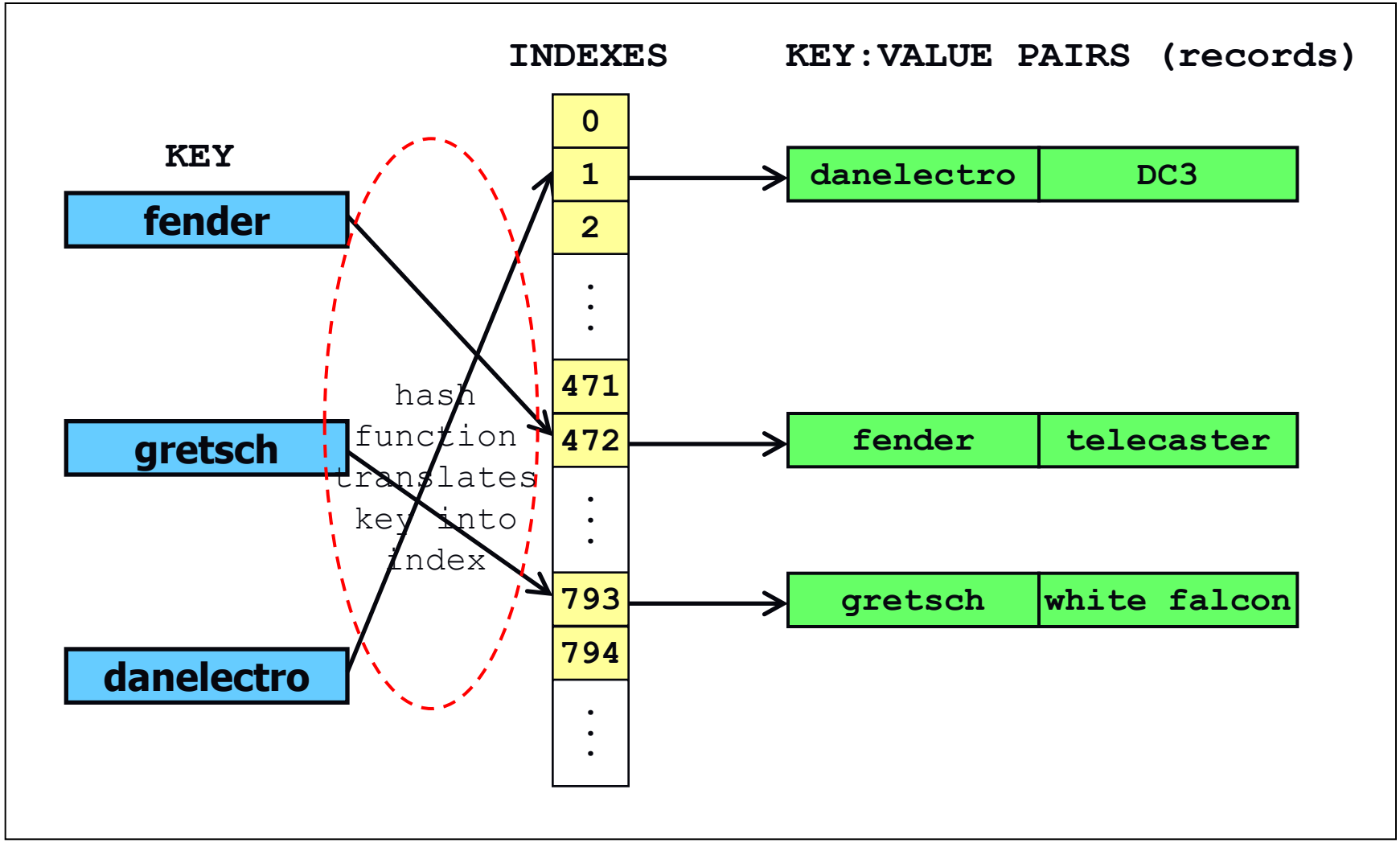
- Like a set in math, e.g. {1, 2, 3, 4}
  - Sets cannot contain duplicate items
- Operations on a Set:
  - Add an item to it
  - Remove an item from it
  - Check if an item is in it
  - Iterate over it (loop based on all items, one-by-one)

Set in Java

- Different ways to implement a set
  - HashSet
    - Faster implementation, but it is unordered
  - TreeSet
    - Slower, but the items are available in order

Map (as a hash table)

- A ***Map*** is a lookup table that takes a ***key*** and returns a ***value***
  - the most common implementation is as a hashtable (hashmap)

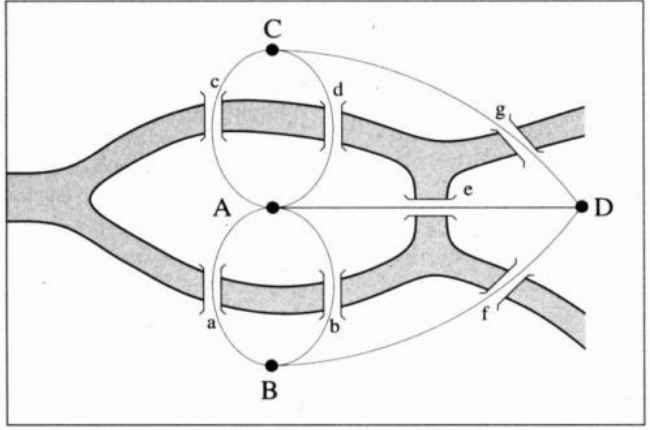


Graphs

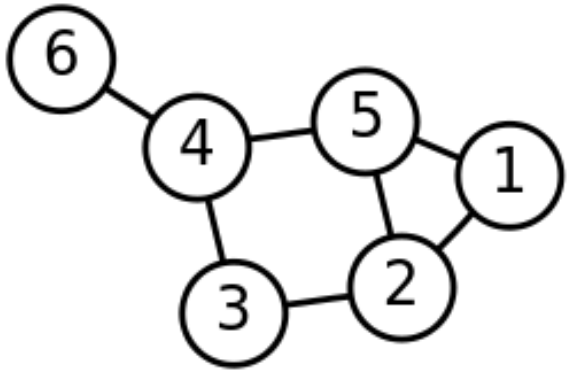
(Still in Chapter 1.4)

Graphs

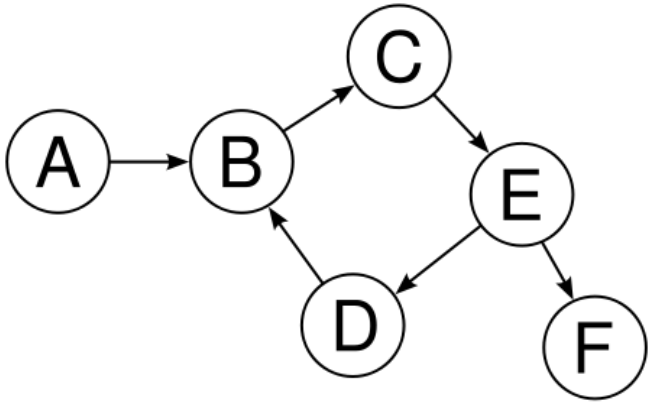
- G = (V, E)
  - V is a set of *vertices*
  - E is a set of *edges*



Motivation: Real world connections



Undirected



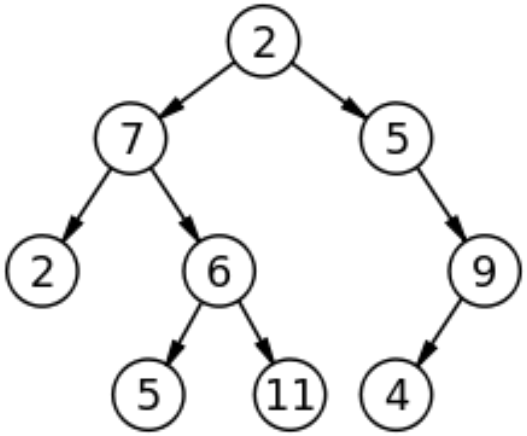
Directed

Some special graphs

- *Connected graph*
  - A graph where there is a path available between any two vertices
- *Cyclic graph*
  - A graph containing at least one cycle
- *Acyclic graph*
  - A graph containing no cycles
- *Tree*
  - Any connected + acyclic graph
- *Complete graph*
  - Every pair of vertices is connected by an edge
- *Weighted graph*
  - Every edge has an associated value

Trees

- Connected, acyclic graphs
  - Usually we think of trees as having a root
- Representing data in a tree can speed up your algorithms in many natural problems

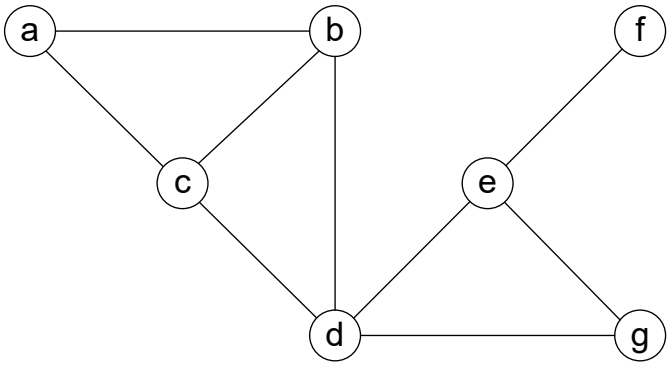


Representation of graphs

- Two common ways to represent graphs:
  - Adjacency matrix
    - |V| x |V| matrix
    - Cell i, j represents an edge from vertex i to j
  - Adjacency lists
    - |V| linked lists – one for each vertex, showing all the neighbours of that vertex

Representation: Adjacency Matrix

- For this graph:

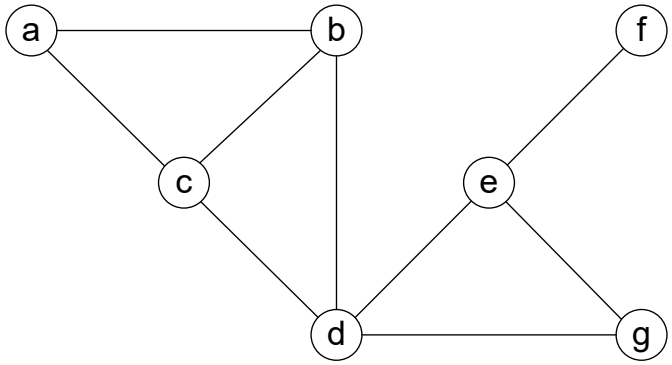


- Adjacency matrix is the following:

	a	b	c	d	e	f	g
a	0	1	1	0	0	0	0
b	1	0	1	1	0	0	0
c	1	1	0	1	0	0	0
d	0	1	1	0	1	0	1
e	0	0	0	1	0	1	1
f	0	0	0	0	1	0	0
g	0	0	0	1	1	0	0

Representation: Adjacency List

- For the same graph:



- Adjacency list is the following:
  - *For vertex a*: b → c
  - *For vertex b*: a → c → d
  - *For vertex c*: a → b → d
  - *For vertex d*: b → c → e → g
  - *For vertex e*: d → f → g
  - *For vertex f*: e
  - *For vertex g*: d → e

Representing Graphs

1. Adjacency matrix
   - Or Weight Matrix for weighted graphs

2. Adjacency lists
   - A list of vertices connected to each vertex

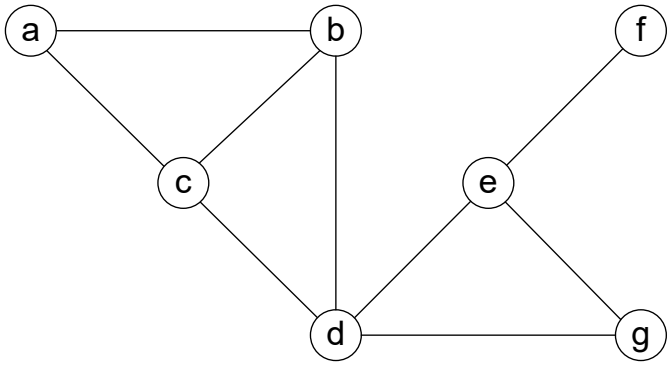
- Which one to use?
  - Depends on the nature of the graph (sparse or not)
  - Depends on the algorithm

Graph Algorithms

(Chapter 3.5)

Graph Traversal

- Many real-world problems require processing of each vertex (or edge) in a graph



  - Routing a message on a network
  - Web crawling
  - Social networking
  - Garbage collection
  - Solving puzzles

Graph Traversal Algorithms

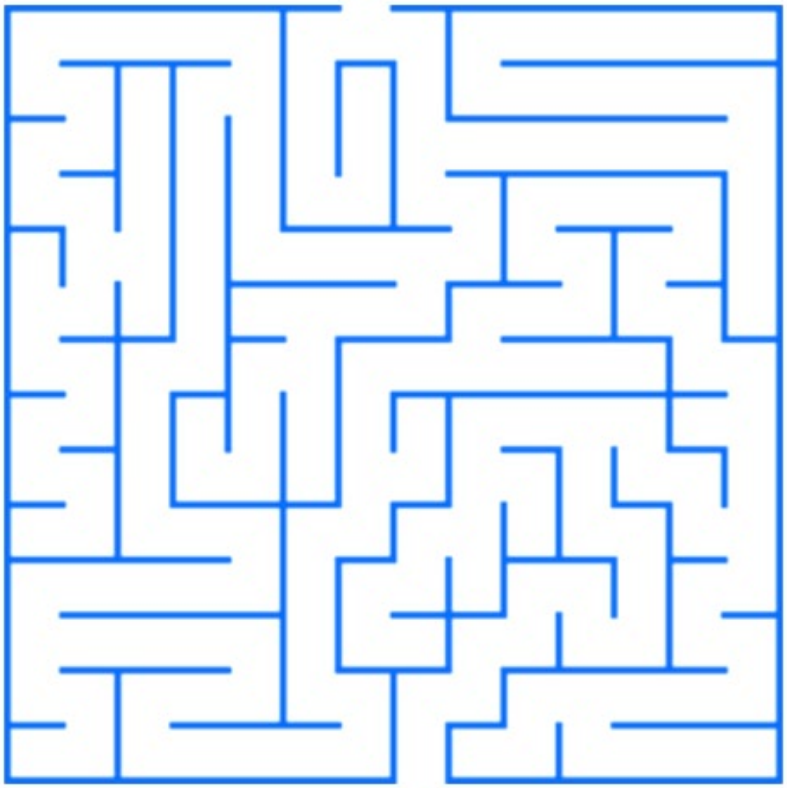
- Graph traversal algorithms give a method for *systematically processing* all vertices

Basic idea: "visit" all the vertices, one at a time, ***marking*** them as we visit them

- Two approaches:
  - Depth-First Search (DFS)
  - Breadth-First Search (BFS)

Depth-first search (DFS)

- Think about how you might try to find your way through a maze



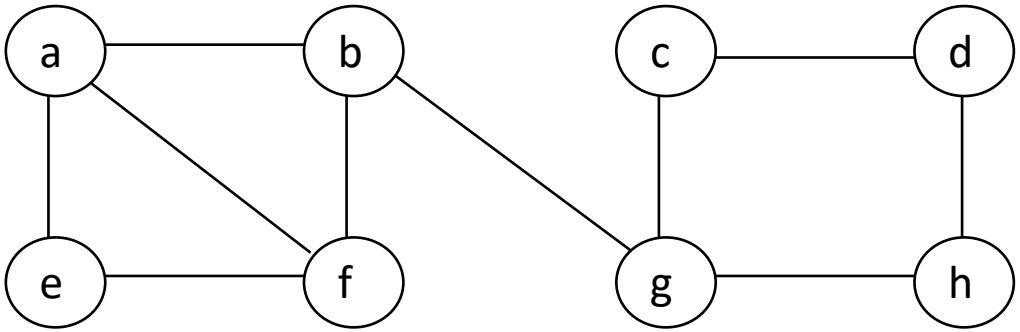
Depth-first search (DFS)

- Visits all vertices by always *moving away* from the last vertex visited (if possible)
  - Backtracks at “dead-ends” (no adjacent, unvisited vertices)

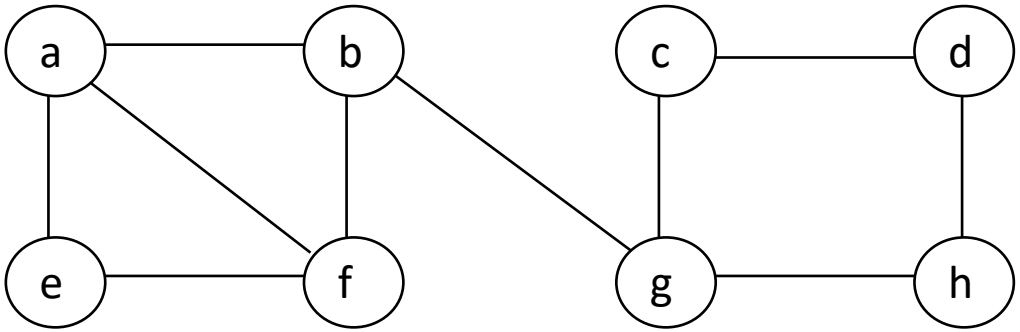
- Implementation often uses a stack of vertices being processed

- Follows a tree-like route throughout the graph

DFS example



DFS example



Backtrack/finish order: e f h d c g b a

DFS order: a b f e g c d h

Some notes on DFS

- To track the progress of the algorithm we use a stack
  - When we make a recursive call, e.g. dfs(v), we push v onto the stack
  - When v is a dead-end (i.e. no more neighbors to visit) it is popped off the stack
- Our convention: break ties for “next neighbor” by using some natural order
- Typical results from running DFS can be:
  - List of vertices in order visited
  - List of vertices in order of “dead-ends” (when popped from stack)
  - DFS Tree – tree containing all the edges that were used to visit nodes
    - Unused edges of G (edges not in DFS tree) are called “back edges”

DFS algorithm

```
Algorithm Depth_First_Search(Graph G)
// Graph G = {V,E}
   initialize visited to false for all vertices
   for each vertex v in V
      if v has not been visited
         dfs_helper(v)

function dfs_helper(Vertex v)
   visit node v
   for each vertex w in V adjacent to v
      if w has not been visited
         dfs_helper(w)
```

- “Visit node v” means doing whatever you need to do at each node

Common uses of DFS

- Find a spanning tree of a graph
- Find a path between two vertices v and u
- Find a path out of a maze
- Determine if a graph has a cycle
- Find all the connected components of a graph
- Search the state-space of problems for a solution (AI)
- Many more!

Efficiency of DFS

- The basic operation is the if statement in dfs_helper():

```
for each vertex w in V adjacent to v
    if w has not been visited
        dfs(w)
```

- Each time dfs_helper() is called, this loop examines all the neighbors of some vertex v … eventually it looks at ALL the neighbors of ALL the vertices
  - Therefore the number of basic operations depends on the data structure used to implement the graph

- Basically we need to visit each element of the data structure exactly once. So the efficiency must be:
  - $O(|V|^2)$ for adjacency matrix
  - $O(|V|+|E|)$ for adjacency lists

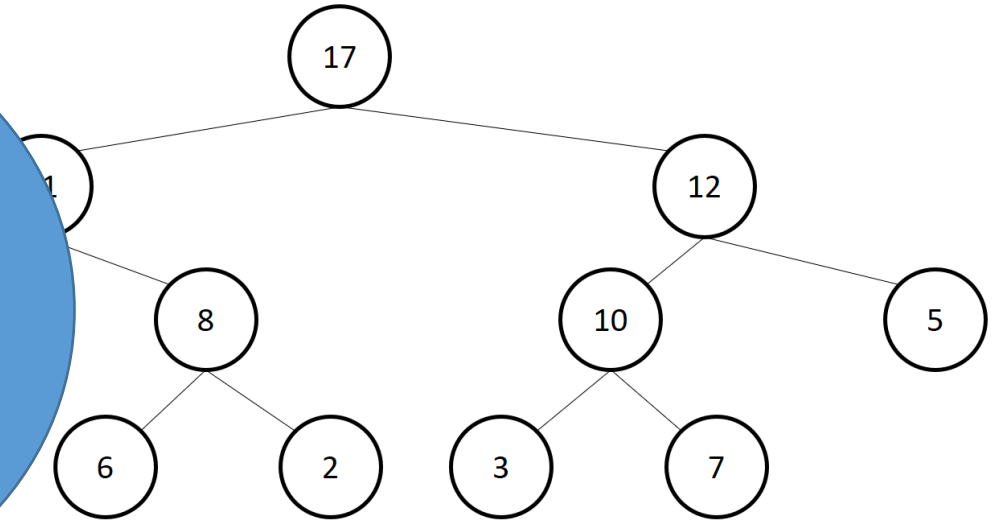
Which is better/worse?

- $O(|V|^2)$
- $O(|V|+|E|)$

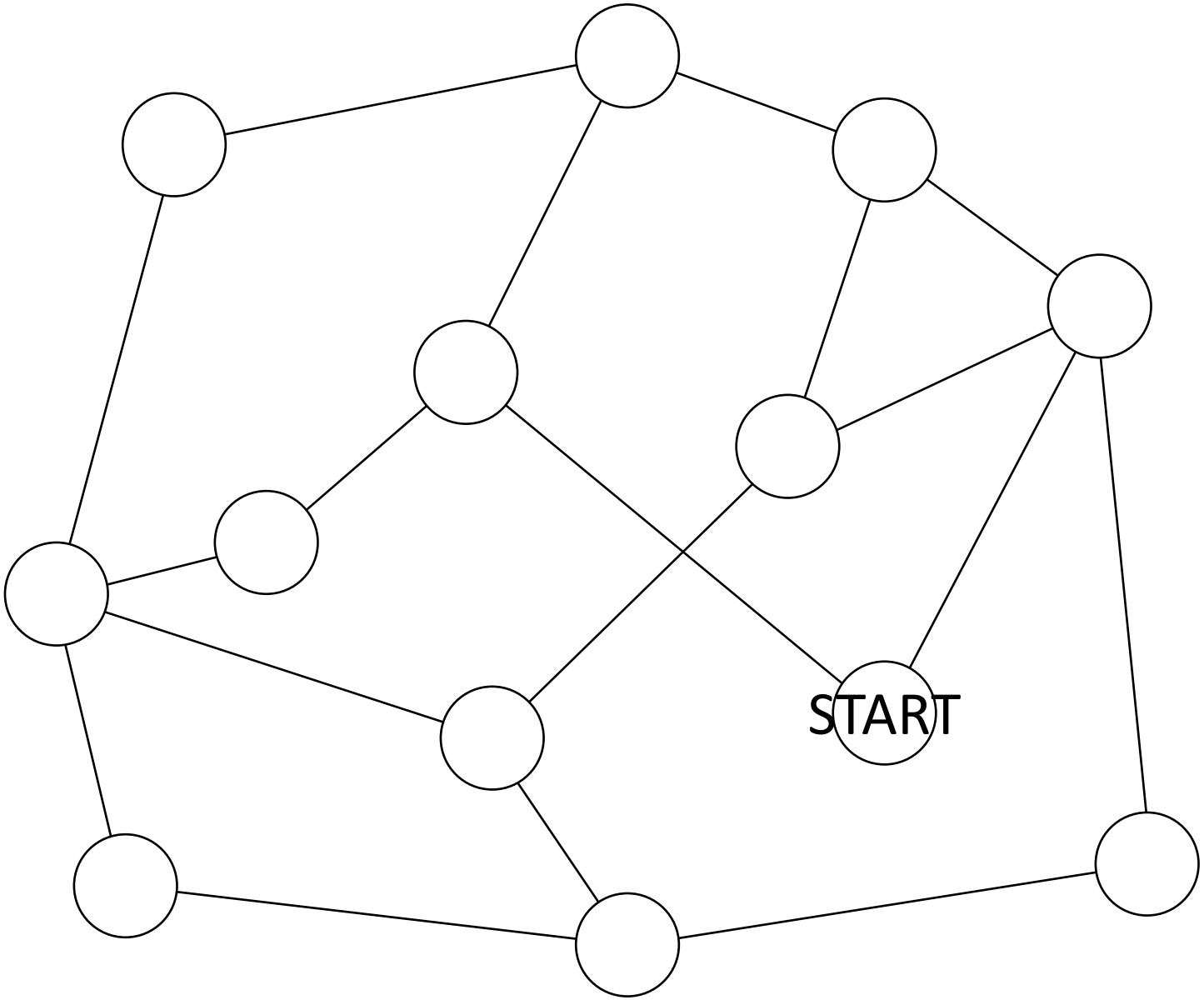
Breadth-first search (BFS)

- Recall the “trick” where we used an array to store a (very specific type of) binary tree (a heap).

Breadth First Search is sorta like this —“level by level” — but it’s about *TRAVERSING A GRAPH*



17 11 12 9 8 10 5 1 4 6 2 3 7


START

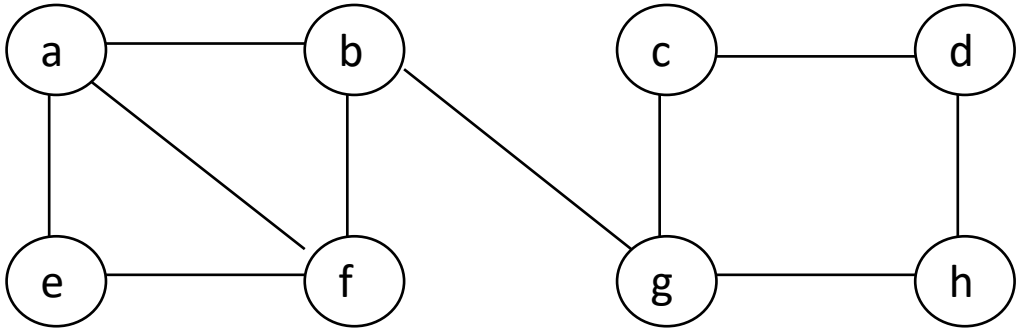
Breadth-first search (BFS)

- Visit all neighbors “the same distance” from starting vertex
  - Visit immediate neighbors first
  - Then the neighbors of all those vertices
  - Etc.

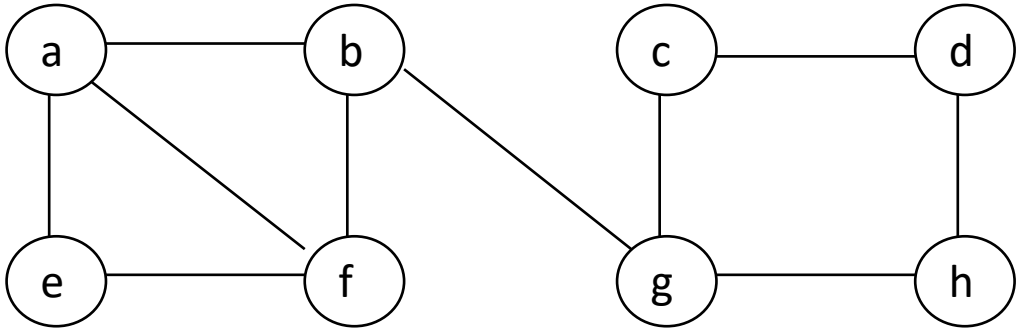
- Instead of a stack, BFS uses a queue

- Follows a tree-like route throughout the graph, but perhaps a different tree than DFS

BFS example



BFS example



BFS order: a b e f g c h d

BFS algorithm

```
Algorithm Breadth_First_Search(Graph G)
// Graph G = {V,E}
   initialize visited to false for all vertices
   for each vertex v in V
      if v has not been visited
         bfs_helper(v)

function bfs_helper(Vertex v)
   visit node v
   initialize a queue Q
   add v to Q
   while Q is not empty
      for each w adjacent to Q.head
         if w has not been visited
            visit node w
            add w to Q
      Q.dequeue()
```

- Uses a queue (FIFO) to determine which vertex to visit next
- Edges that are in G but not in the resulting BFS tree are called "cross-edges"

Notes on BFS

- Same efficiency as DFS:
  - Adjacency matrix: $O(|V|^2)$
  - Adjacency list: $O(|V|+|E|)$

- Yields just one ordering of vertices (order added/deleted from queue is the same)
  - Whereas with DFS, the order that vertices are *visited* may be different from the order they get *finished* (become dead-ends)

BFS applications

- Really the same as DFS
- Sometimes one or the other may be better for specific problems

Some problems solvable with graph traversal

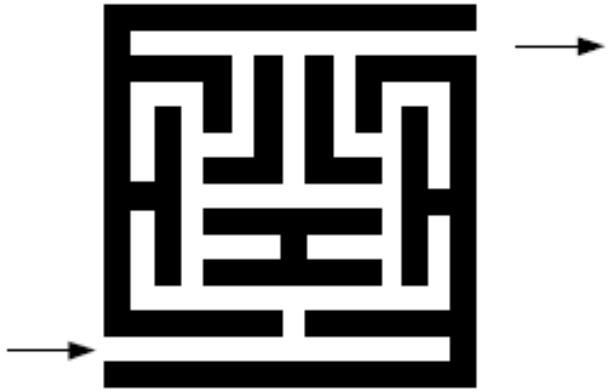
Problem 1: Spanning Tree

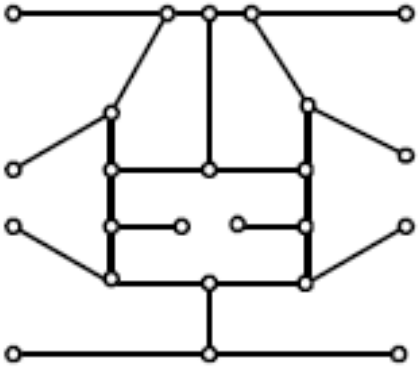
- Given a connected graph G, find a spanning tree T
  - This is a straight-up application of BFS (or DFS)
  - Build a new graph (the spanning tree) as we go
    - Initialize a new graph T
    - Each time we visit a vertex, add the edge we used to T

- BFS or DFS?
  - BFS usually gives shorter paths between vertices

Problem 2: Solving a Maze

- Represent maze as a graph
  - Nodes for start, finish, intersections, and dead-ends
  - Find a path from *start* to *finish*
- BFS or DFS?
  - If interested in end-result:
    - BFS will find the shortest total path
  - If actually walking while solving:
    - DFS tends to result in less actual walking
    - BFS backtracks to parent nodes too often





Problem 3: Shortest Path

- Find the shortest path between two vertices u and w
- BFS or DFS?

Problem 3: Shortest Path

- Find the shortest path between two vertices u and w
- BFS or DFS?
  - BFS will find a shortest path
  - DFS will find a path – maybe not the shortest one

- Idea of algorithm (and why it works):
  - First, use bfs(u) to create a spanning tree T with u as the root. Note that all paths that appear in T are the shortest paths from u to their respective vertices
  - Then, use DFS on T to find a path from u to w (as in the maze problem)

Problem 4: Determine Connectivity

- Determine if a graph is connected
- BFS or DFS?
  - Either will work
  - Think about this yourself!
  - What modification(s) do you need to make to the algorithm to answer this question?

Graph Algorithms: Binary tree traversal

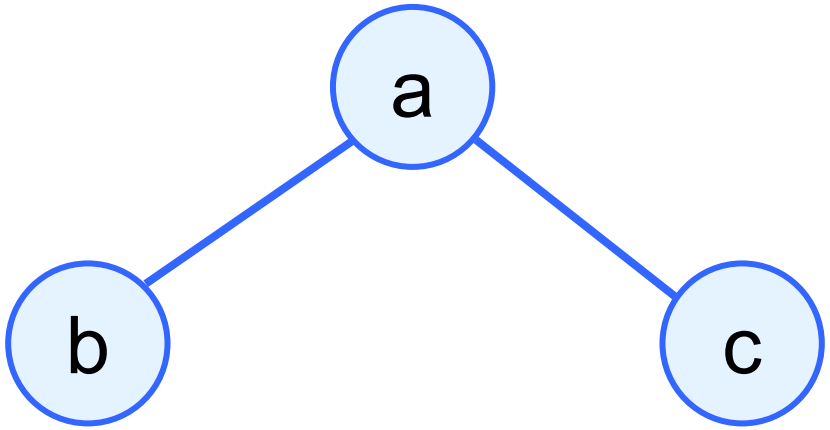
Textbook: Chapter 5.3

Tree traversal

- Traversing a tree means to visit all of the nodes of the tree
- We've already seen DFS and BFS (for graphs)
- Here are a few traversals specific to *binary trees*:
  - Preorder – root *before* the children
  - Inorder – root *between* the children
  - Postorder – root *after* the children

Preorder traversal

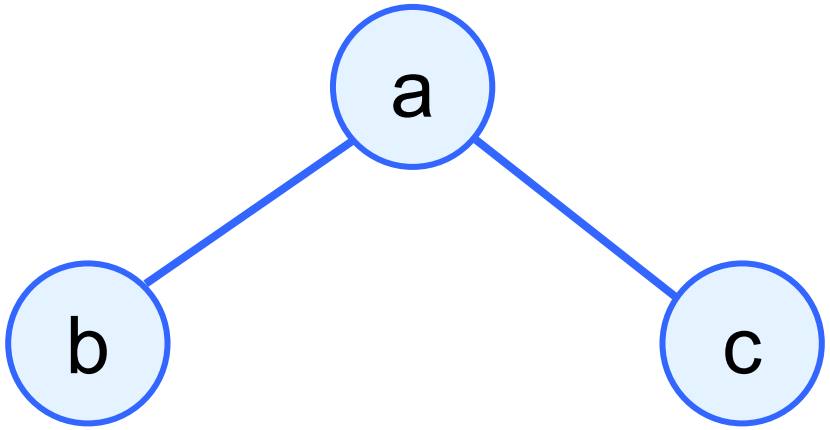
1. Visit the root
2. Traverse the left subtree
3. Traverse the right subtree



Preorder traversal is: a b c

Inorder traversal

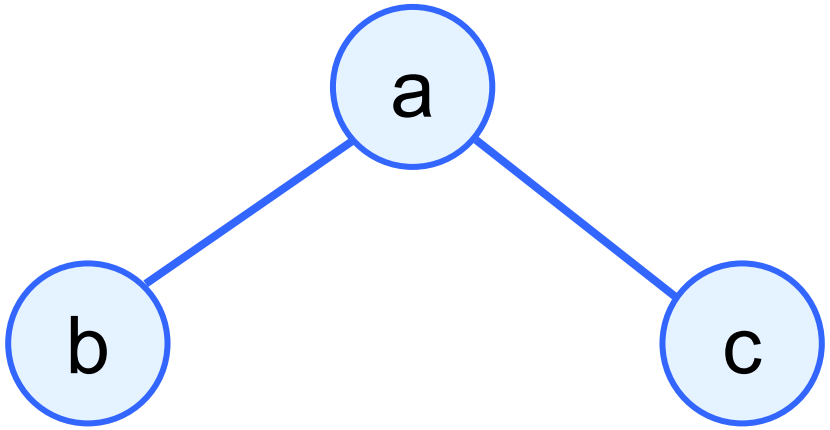
1. Traverse the left subtree
2. Visit the root
3. Traverse the right subtree



Inorder traversal is: b a c

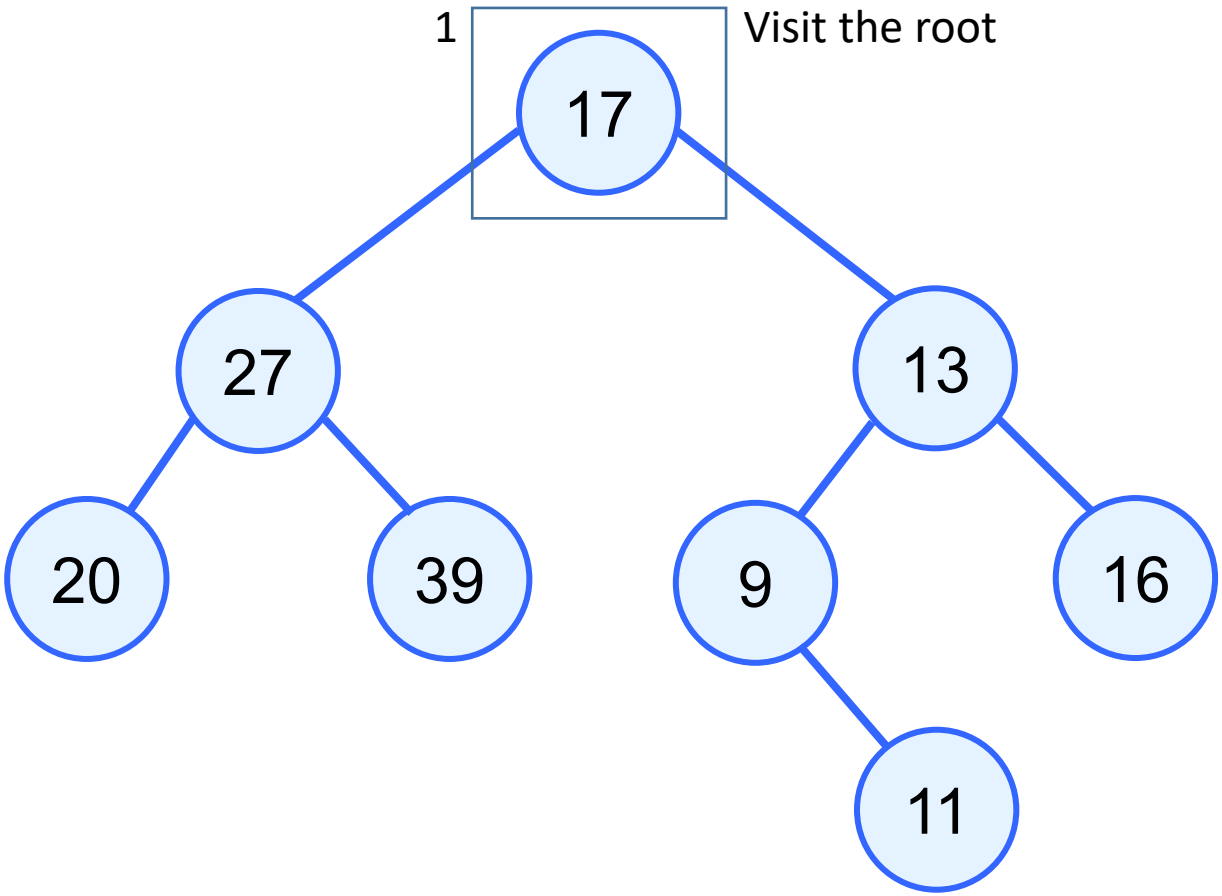
Postorder traversal

1. Traverse the left subtree
2. Traverse the right subtree
3. Visit the root



Postorder traversal is: b c a

Preorder example



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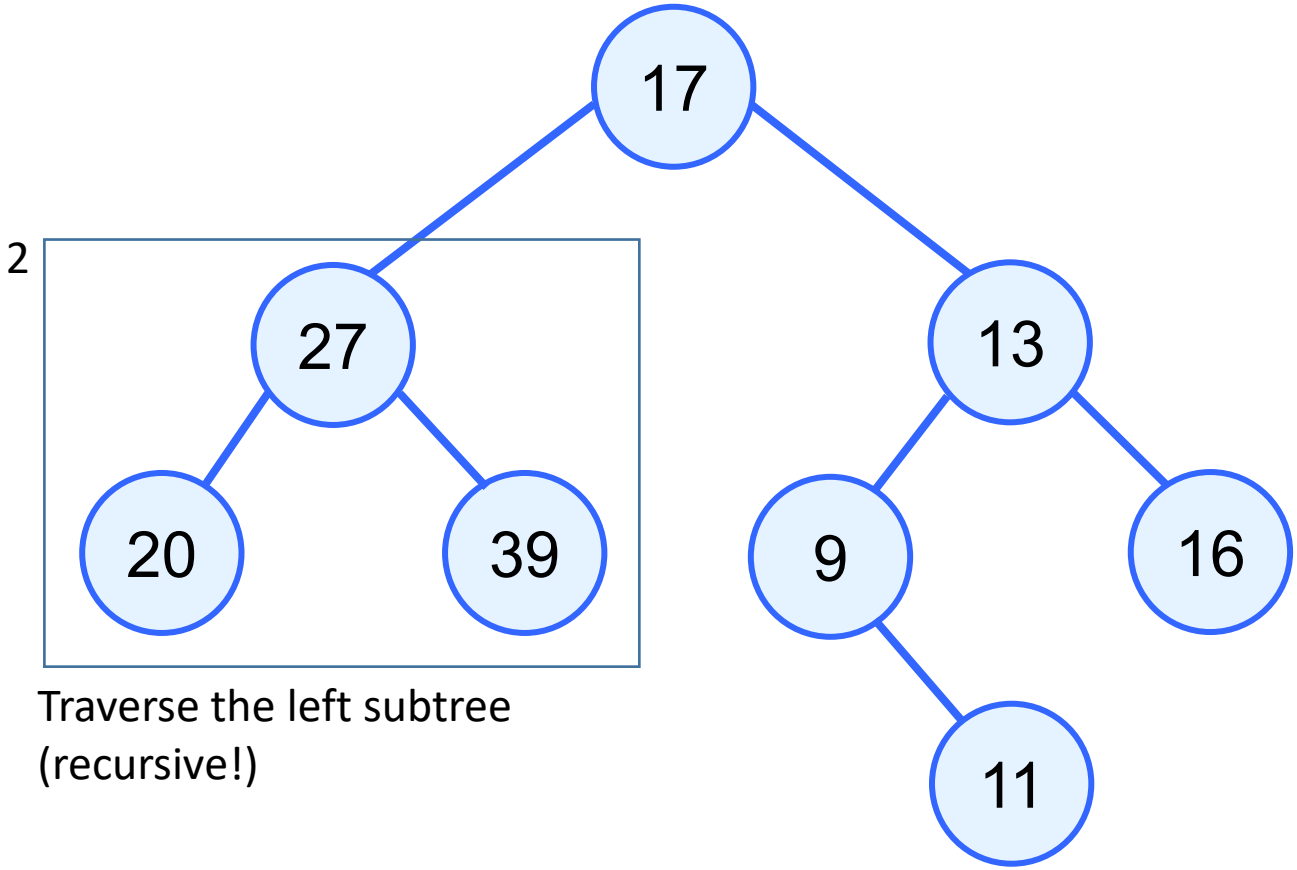
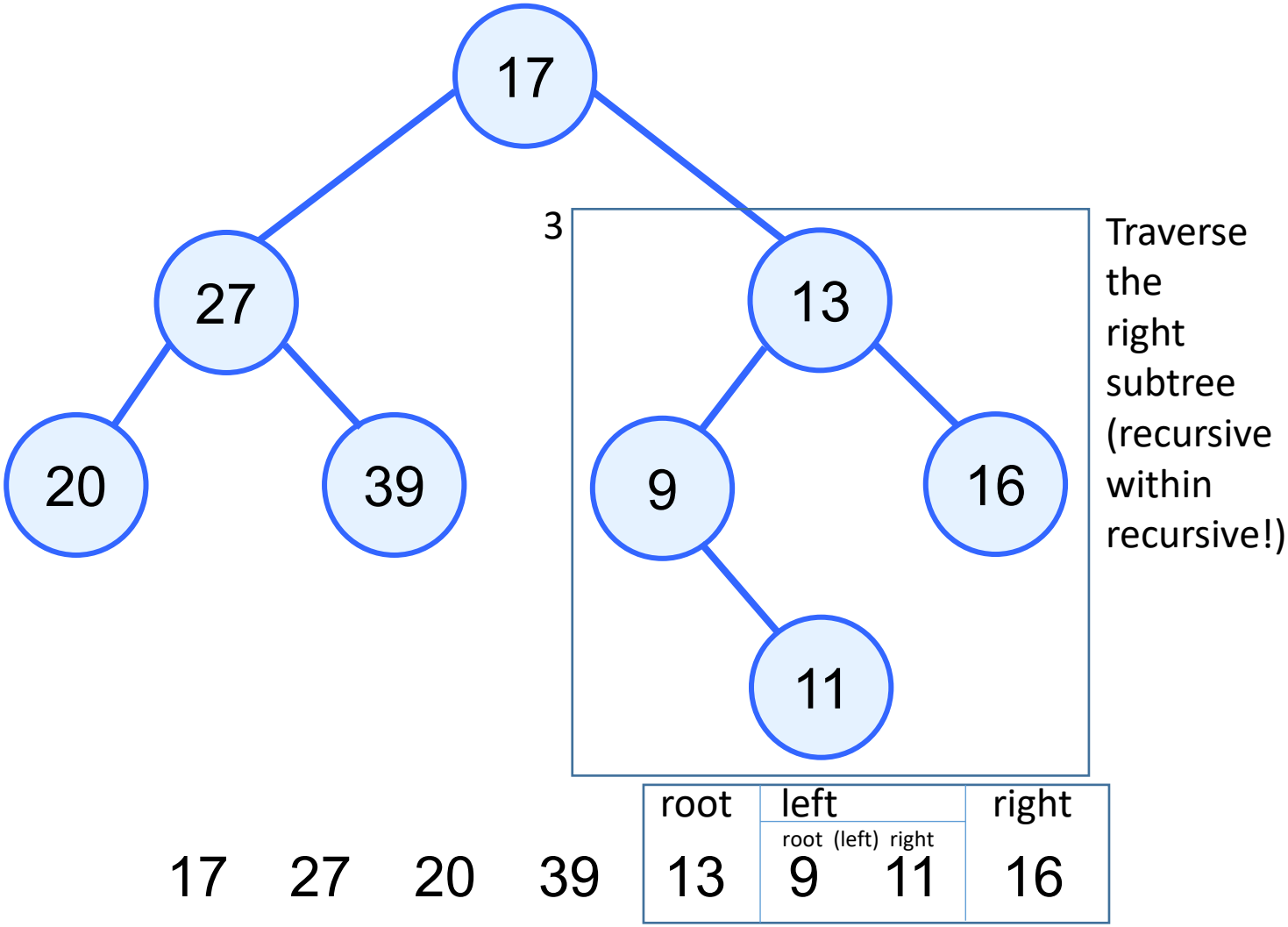
Preorder example



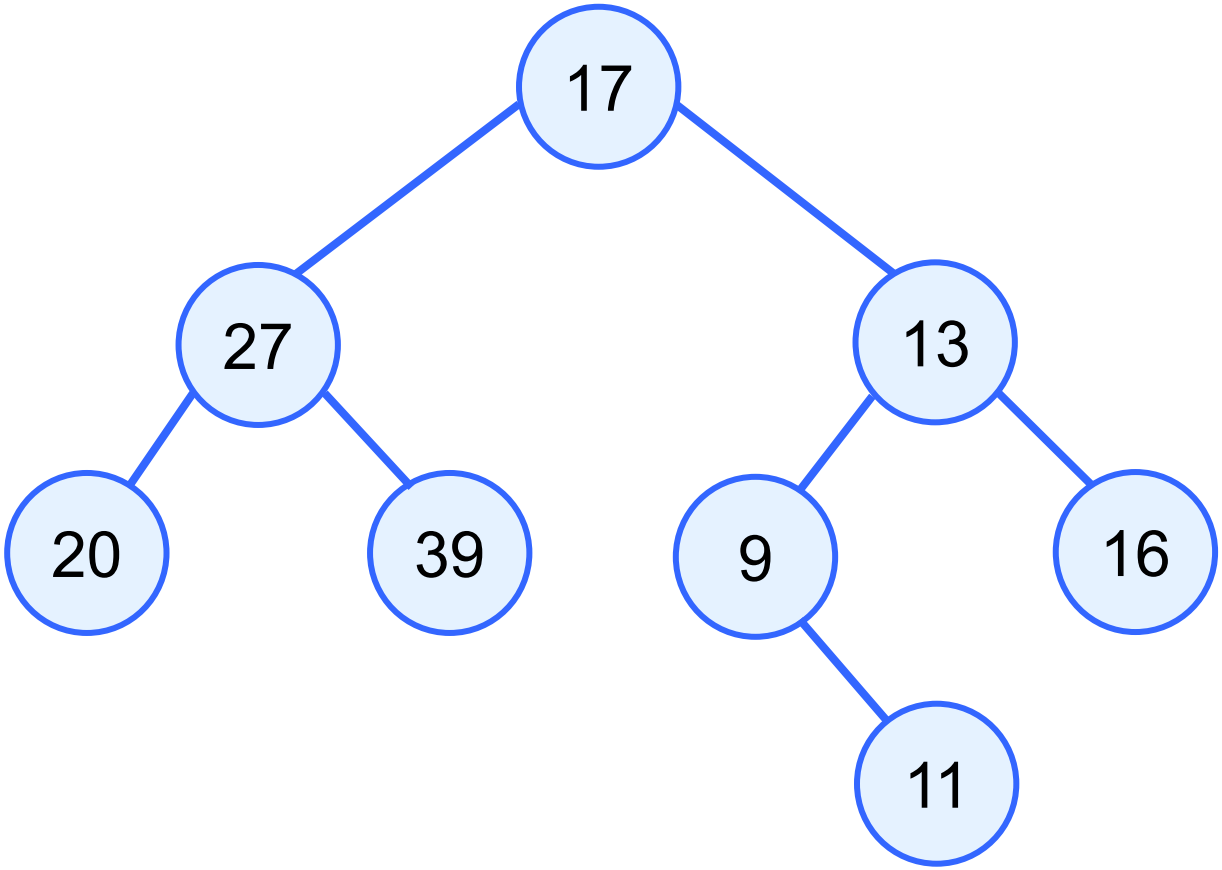
Traverse the left subtree
(recursive!)

	root	left	right
17	27	20	39

Preorder example



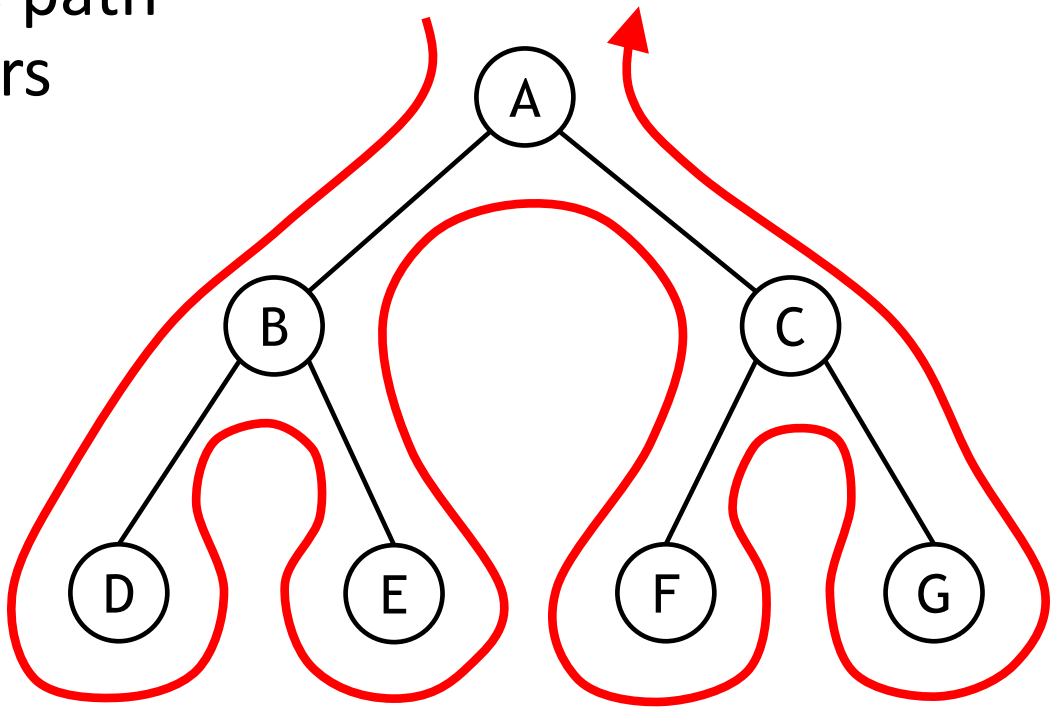
Preorder example



17 27 20 39 13 9 11 16

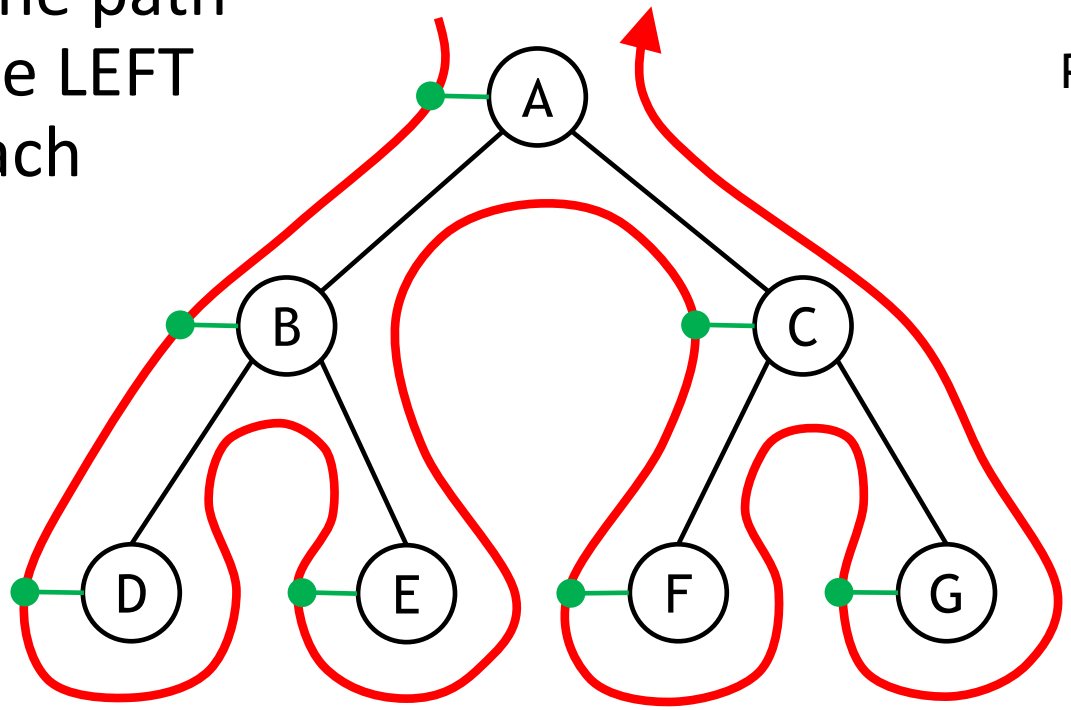
Another way to think about it

- Consider this path that meanders past all of the nodes
- ALL three traversals follow this path!



Preorder traversal

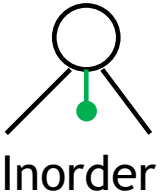
- Is when the path passes the LEFT side of each node

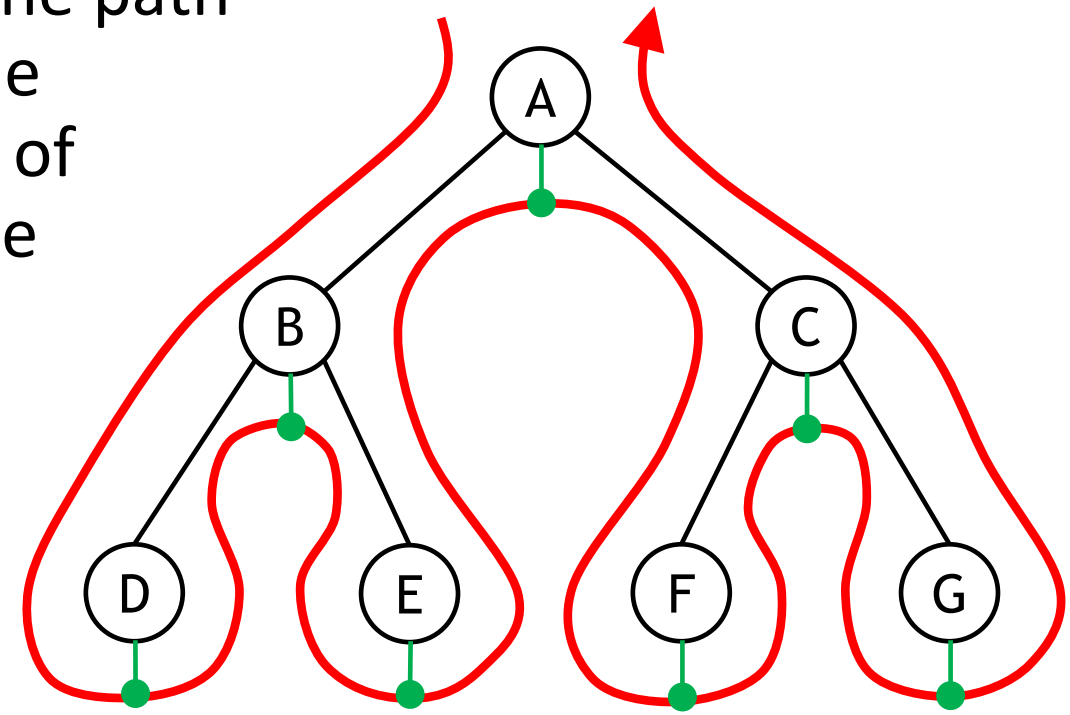


Preorder: A B D E C F G

Inorder traversal

- Is when the path passes the BOTTOM of each node

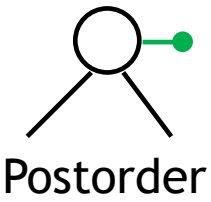


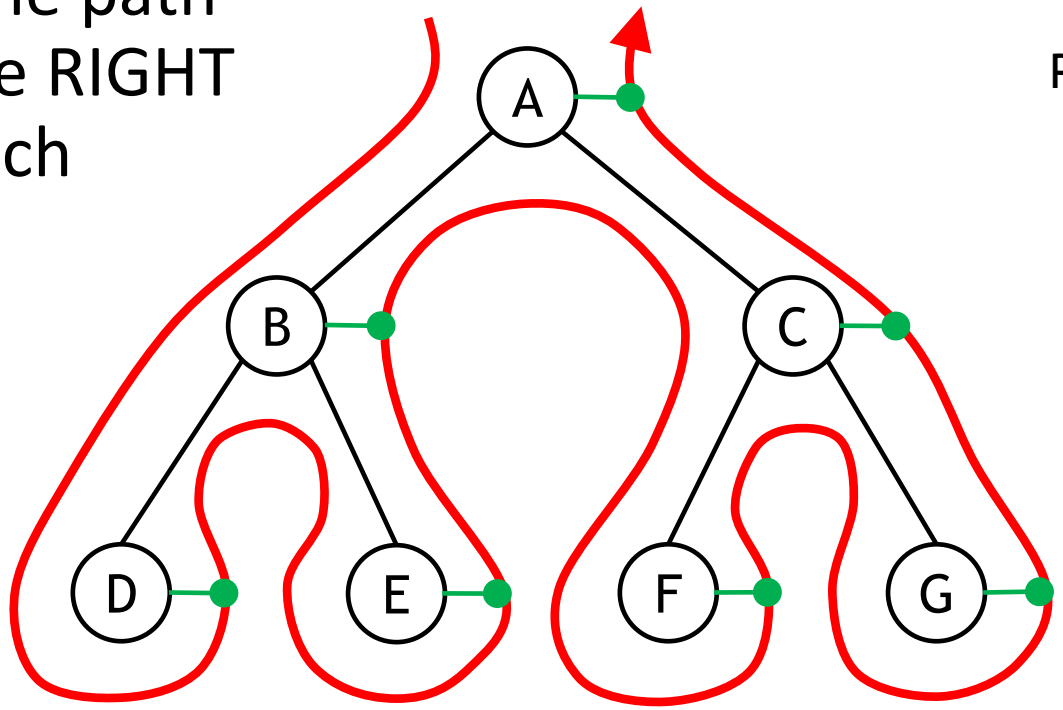


Inorder: D B E A F C G

Postorder traversal

- Is when the path passes the RIGHT side of each node





Postorder: D E B F G C A

Pseudocode

```
Algorithm inOrder(Node N)
if N != null
   inOrder(N.leftChild)
   Print N.value
   inOrder(N.rightChild)
```

```
Algorithm preOrder(Node N)
if N != null
   Print N.value
   preOrder(N.leftChild)
   preOrder(N.rightChild)
```

```
Algorithm postOrder(Node N)
if N != null
   postOrder(N.leftChild)
   postOrder(N.rightChild)
   Print N.value
```

What if I told you

Preorder → A B D E C F G

Inorder → D B E A F C G

Fun facts about pre/in/postorder

- Given pre + in, you can reconstruct the tree
  - (and also determine postorder)
- Given post + in, you can reconstruct the tree
  - (and also determine preorder)
- Given pre + post, you can only *sometimes* reconstruct the tree
  - For you to ponder: under what condition(s)?

DFS algorithm

```
Algorithm Depth_First_Search(Graph G)
// Graph G = {V,E}
   initialize visited to false for all vertices
   for each vertex v in V
      if v has not been visited
         dfs_helper(v)

function dfs_helper(Vertex v)
   visit node v
   for each vertex w in V adjacent to v
      if w has not been visited
         dfs_helper(w)
```

- “Visit node v” means doing whatever you need to do at each node

BFS algorithm

```
Algorithm Breadth_First_Search(Graph G)
// Graph G = {V,E}
   initialize visited to false for all vertices
   for each vertex v in V
      if v has not been visited
         bfs_helper(v)

function bfs_helper(Vertex v)
   visit node v
   initialize a queue Q
   add v to Q
   while Q is not empty
      for each w adjacent to Q.head
         if w has not been visited
            visit node w
            add w to Q
      Q.dequeue()
```

- Uses a queue (FIFO) to determine which vertex to visit next
- Edges that are in G but not in the resulting BFS tree are called "cross-edges"